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Major – Economics

Eco – 301

HOMEWORK – 1

Ans 1

$U(x,y) = x^a y^{1-a}$ s.t. $0 \leq a \leq 1$, x – basketball game. y – football game.

Budget constraint: $M = P_x \cdot x + P_y \cdot y$ where: M = CD's income

P_x, P_y = ticket prices

To find the demand function for x & y, we need to maximize $U(x,y) = x^a y^{1-a}$ subject to $M = P_x \cdot x + P_y \cdot y$.

Using Lagrange Multipliers,

$$L(x,y) = x^a y^{1-a} + \lambda (M - P_x \cdot x - P_y \cdot y)$$

Partially derivating the equation with x, then y and then λ.

$$\frac{\partial L}{\partial x} = y^{1-a} \cdot a x^{a-1} - \lambda P_x = 0 \quad -(1)$$

$$\frac{\partial L}{\partial y} = x^a \cdot (1-a) y^{-a} - \lambda P_y = 0 \quad -(2)$$

$$\frac{\partial L}{\partial \lambda} = M - P_x \cdot x - \lambda P_y \cdot y = 0 \quad -(3)$$

From eq^n (1),

$$y^{1-a} \cdot a x^{a-1} = \lambda P_x$$

$$\boxed{a \cdot \frac{y}{y^a} \cdot \frac{x^a}{x} = \lambda P_x} \quad -(4)$$

From eq^n (2),

$$x^a (1-a) y^{-a} = \lambda P_y$$

$$\boxed{(1-a) \cdot \frac{x^a}{y^a} = \lambda P_y} \quad -(5)$$

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Dividing eqn ④ by ⑤, To find the utility maximization condition $MRS = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}$,

$$\frac{\cancel{\lambda} P_x}{\cancel{\lambda} P_y} = \frac{a \cdot \frac{y}{y^a} \cdot \frac{x^a}{x}}{(1-a) \cdot \frac{x^a}{y^a}} = \frac{a \cdot y \cdot \cancel{x^a}}{\cancel{y^a} \cdot x} \times \frac{\cancel{y^a}}{(1-a) \cdot \cancel{x^a}}$$

$$\frac{P_x}{P_y} = \frac{ay}{(1-a)x}$$

$$\therefore \boxed{x = \frac{ay}{(1-a)} \times \frac{P_y}{P_x}} \quad -⑥$$

Now, substituting this value of x in the budget constraint eqn: $M = P_x \cdot x + P_y \cdot y$

$$\Rightarrow M = \cancel{P_x} \cdot \left(\frac{a \cdot y}{(1-a)} \times \frac{P_y}{\cancel{P_x}} \right) + P_y \cdot y$$

$$M = y \left[\frac{a P_y}{(1-a)} + P_y \right]$$

$$M = y \left[\frac{a P_y + (1-a) P_y}{(1-a)} \right]$$

$$y = \frac{M(1-a)}{a P_y + (1-a) P_y}$$

$$y = \frac{M(1-a)}{\cancel{a P_y} + P_y - \cancel{a P_y}}$$

$$\boxed{y = \frac{M(1-a)}{P_y}}$$ → Demand function of y.

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Substituting the demand function of y in eq^n ⑥,

$$x = \frac{ay}{(1-a)} \times \frac{P_y}{P_x}$$

$$x = \frac{a}{\cancel{(1-a)}} \cdot \frac{M\cancel{(1-a)}}{\cancel{P_y}} \times \frac{\cancel{P_y}}{P_x}$$

$$\boxed{x \Rightarrow a \cdot \frac{M}{P_x}}$$ → Demand function of x.

So, Mr CD's demand function for base ball tickets is:

$$\boxed{x = a \cdot \frac{M}{P_x}}$$

and for football tickets are:

$$\boxed{y = (1-a)\frac{M}{P_y}}$$

This shows that the goods are normal, as for each good, as income M goes up, the quantity demanded ~~goo~~ goes up. Also, the goods are neither complements nor the substitutes of one another. Since the demand function of any good is not dependent upon the price of the other, the fact that the price of one rises or falls has no effect on the demand for the other.

(b) $U(x,y) = x^a y^{1-a}$

When $a=0$, Mr. CD doesnot get any utility from x, all he cares about is y and thus spends his entire income on y. If $a=1$, he cares only about x and not about y, thus spends his entire income on x. When $a = 1/2$, he spends his income equally between x and y, no matter what their prices might be.

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Ans 2

(a) $x \rightarrow$ cookies
$y \rightarrow$ bananas

Budget line for the first three cookies (free):

General Eq^n: $P_x \cdot x + P_y \cdot y = M$ where M = Income.

Here, $0 \cdot x + 1 \cdot y = 5$ [Price of Bananas $P_y = \$1$, ~~Price of~~ Income $= \$5$]

$y = 5$. —①

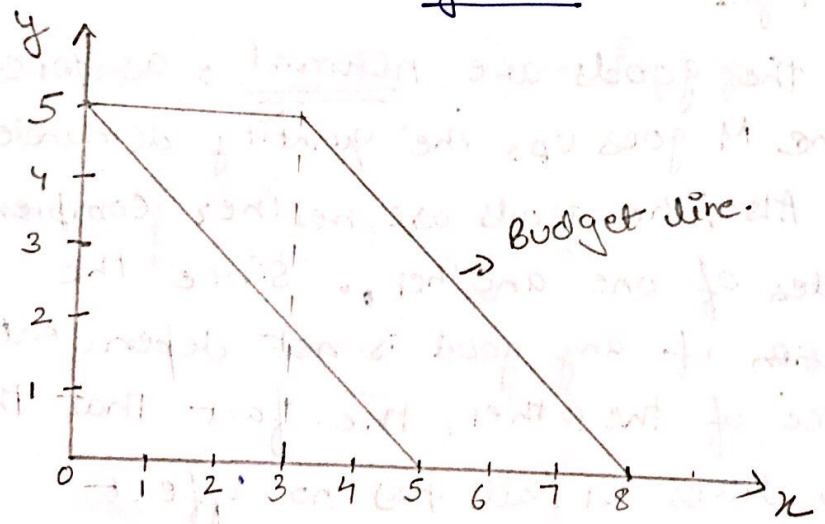
Budget line after the first three cookies:

$1 \cdot x + 1 \cdot y = 5$ [Since price of cookies $= \$1$ now]

$x + y = 5$ —②

Considering the two eq^{n}s, drawing the budget line:

Figure 1



Since the first equation, $y = 5$ is a line parallel to the x axis, the budget line will initially be a horizontal line at $y = 5$, till $x = 3$, which is the point till when the cookies are free.

Then from the second equation $x + y = 5$,

$\frac{dy}{dx} = -1$. So, we get a straight line with a slope of -1.

After $x = 3$, the budget line will be a line with the slope -1, parallel to the original straight line.

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(b) OPTIMAL CONSUMPTION POINT:

The optimal consumption point, we must find the MRS_{xy} from the utility function and solve it with the equation of the budget line.

If $u(x,y) = x(y+3)$, the tangency condition is

$$MRS = \frac{p_1}{p_2}$$

Or $$MRS = \frac{MU_x}{MU_y} = \frac{p_1}{p_2}$$

$$MU_x = \frac{\partial u}{\partial x} = y+3$$

$$MU_y = \frac{\partial u}{\partial y} = x$$

$$\frac{MU_x}{MU_y} = \frac{y+3}{x} = \frac{P_x}{P_y}$$

$$\frac{y+3}{x} = \frac{1}{1} \quad \left[\begin{array}{l} P_x = \$1 \\ P_y = \$1 \end{array}\right]$$

$$y+3 = x$$

$$\therefore \boxed{x - y = 3} \quad -(A)$$

The initial budget line equation was:

$$x+y = 5,$$

but after considering $y = 5$,

We can see in Figure 1

that at $y = 5$, $x = 3$

$\therefore$ The new budget line equation becomes $\boxed{x+y = 8}$, since slope is same, i.e. $\underline{\underline{-1}}$.

(B)

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Solving Ⓐ & Ⓑ,

$$\begin{array}{rl} x - \cancel{y} &= 3 \\ + \quad x + \cancel{y} &= 8 \\ \hline 2x &= 11 \end{array}$$

$$\boxed{x = 5.5}$$

Substituting $x = 5.5$ in Ⓐ,

$$5.5 - y = 3$$

$$\boxed{y = 2.5}$$

$\therefore$ The optimal consumption point is:

$$(x^*, y^*) = (5.5, 2.5).$$

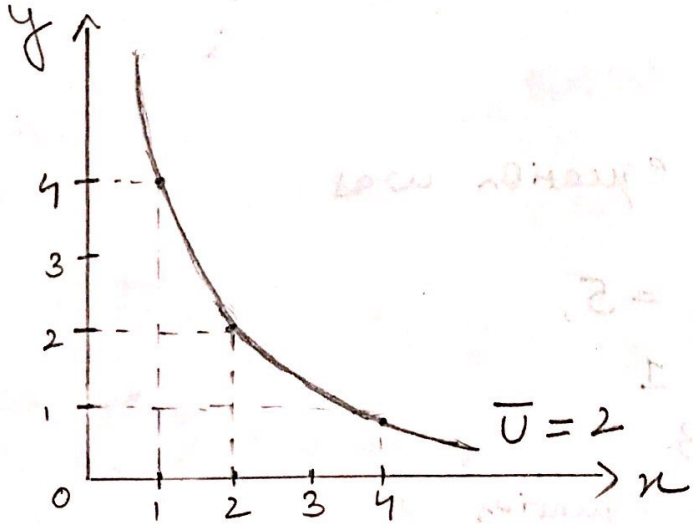
Ans 3.

$$U(x, y) = (x \cdot y)^{1/2}$$

(a) at $u = 2$

$$2 = (xy)^{1/2}$$

$$4 = x \cdot y$$



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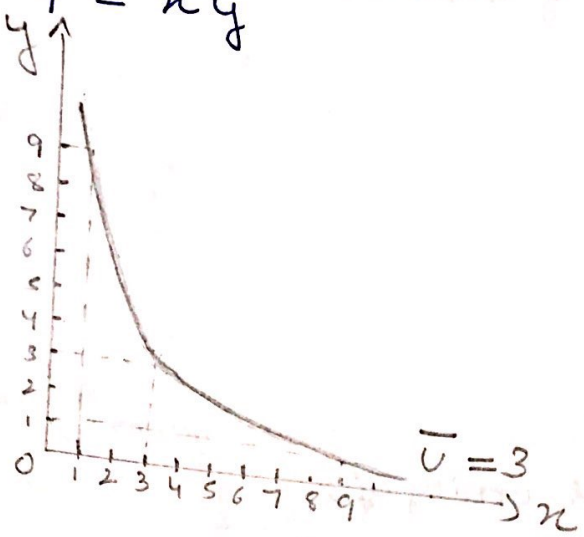
at $U = 3$

$3 = (xy)^{1/2}$

$9 = xy$



(b) Since the indifference curves are bowed towards the origin, they do obey the assumption of diminishing MRS.

$U = (xy)^{1/2}$

$$MU_x = \frac{\partial U}{\partial x} = y^{1/2} \cdot \frac{1}{2} x^{-1/2}$$
$$= \frac{1}{2}\left(\frac{y}{x}\right)^{1/2}$$

$$MU_y = \frac{\partial U}{\partial y} = x^{1/2} \cdot \frac{1}{2} y^{-1/2}$$
$$= \frac{1}{2}\left(\frac{x}{y}\right)^{1/2}$$

From this we can see that both MU_x & MU_y are positive.

$$MRS_{xy} = \frac{MU_x}{MU_y} = \frac{\cancel{1/2}\,(y/x)^{1/2}}{\cancel{1/2}\,(x/y)^{1/2}}$$
$$= \frac{y^{1/2}}{x^{1/2}} \times \frac{y^{1/2}}{x^{1/2}}$$

$$\boxed{MRS_{xy} \Rightarrow \frac{y}{x}}$$

So, as x, increases, the denominator gets bigger and MRS decreases. As X increases, Y decreases and the numerator gets smaller so MRS, decreases. Both these effects work,

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So ~~as~~ that as X increases MRS decreases.
Therefore, the law of diminishing marginal ~~utility~~ ~~is applicable here~~. rate of substitution holds.

(c) $U(x, y) = (xy)^{1/2}$

$x = 4, y = 9;$

$$MU_x = \frac{\partial U}{\partial x} = \frac{1}{2}\left(\frac{y}{x}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{9}{4}\right)^{1/2}$$
$$= \frac{1}{2} \times \frac{3}{2}$$

$MU_x \Rightarrow \boxed{\frac{3}{4}}$ → Marginal utility of x

$$MU_y = \frac{\partial U}{\partial y} = \frac{1}{2}\left(\frac{x}{y}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{4}{9}\right)^{1/2}$$
$$= \frac{1}{2} \times \frac{2}{3}$$

$MU_y \Rightarrow \boxed{\frac{1}{3}}$ → Marginal utility of y.

$$MRS_{xy} = \frac{MU_x}{MU_y}$$
$$= \frac{3}{4} \div \frac{1}{3}$$
$$= \frac{3}{4} \times 3$$

$\boxed{MRS_{xy} \Rightarrow \frac{9}{4}}$ → Marginal rate of substitution of y for x.

Now if $x = 6$ from $x = 4$,
since utility still remains same,
Initial utility,

$U = (xy)^{1/2}$ where $x = 4$, $y = 9$

$U = (4 \times 9)^{1/2} = (36)^{1/2}$

$\boxed{U = 4}$

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Now $x=6$, $U=4$, to find new 'y'.

$$U = (xy)^{1/2}$$
$$4 = (6y)^{1/2}$$
$$16 = 6y$$
$$\frac{16}{6} = y$$
$$\boxed{\frac{4}{3} = y}$$

New marginal utility of x, at $x=6$, $y=\frac{4}{3}$

$$MU_x' = \frac{1}{2}\left(\frac{y}{x}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{4/3}{6}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{4}{18}\right)^{1/2}$$
$$= \frac{1}{2} \times \frac{2}{3\sqrt{2}}$$
$$= \frac{1}{3\sqrt{2}}$$

New marginal utility of y, at $x=6$, $y=\frac{4}{3}$

$$MU_y' = \frac{1}{2}\left(\frac{x}{y}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{6}{4/3}\right)^{1/2}$$
$$= \frac{1}{2}\left(\frac{18}{4}\right)^{1/2}$$
$$= \frac{1}{2} \times \frac{3\sqrt{2}}{2}$$
$$\Rightarrow \frac{3\sqrt{2}}{4}$$

So, new $MRS_{xy}' = \frac{MU_x'}{MU_y'}$

$$= \frac{1}{3\sqrt{2}} \times \frac{4}{3\sqrt{2}}$$
$$= \frac{4}{9 \times 2} \Rightarrow \boxed{\frac{2}{9}} \rightarrow \text{New } MRS_{xy}$$

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(d) $V(x,y) = [U(x,y)]^2$
$= [(xy)^{1/2}]^2$
$\Rightarrow xy$

$MU_x = \frac{\partial U}{\partial x} = y$

$MU_y = \frac{\partial U}{\partial y} = x$

$MRS_{xy} = \frac{MU_x}{MU_y} = \frac{y}{x}$

at $x = 4,\ y = 9$

$\boxed{MRS_{xy} = \frac{9}{4}}$

$\therefore$ The MRS_{xy} remains the same as that of U ~~before~~, since when a function is increasingly transformed, MRS remains same.

(e) $W(x,y) = x(y)^{1/2}$

For this f^n to be an increasing transformation of U, the ~~Mrs~~ MRS_{xy} of W & U should be ~~equal~~. Equal ~~and~~ ~~multiples~~.

MRS_{xy} of W,

$W(x,y) = x(y)^{1/2}$

$MU_x = \frac{\partial U}{\partial x} = y^{1/2}$

$MU_y = \frac{\partial U}{\partial y} = x \times \frac{1}{2}(y)^{-1/2}$
$= \frac{1}{2}\frac{x}{y^{1/2}}$

~~MRS~~ $MRS_{xy} = \frac{MU_x}{MU_y} = \frac{y^{1/2}}{\frac{1}{2}\frac{x}{y^{1/2}}} = y^{1/2} \times \frac{2y^{1/2}}{x}$

$\Rightarrow \boxed{\frac{2y}{x}}$

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The MRS $_{xy}$ of $U = \frac{y}{x}$ {as calculated in (b)}

Since $\frac{y}{x} \neq \frac{2y}{x}$

i.e. MRS $_{xy}$ of $U \neq$ MRS $_{xy}$ of W

$\therefore$ The f^n $W(x,y) = x(y)^{1/2}$ is <u>not</u> an increasing transformation of <u>U</u>.

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